

DSC 335 Project Spring 2025

Overview

- Find a data set with multiple predictor variables and one quantitative or categorical response variable
- Propose a main effects, first-order, Bayesian regression model with at least two predictor variables (no interaction or higher order terms).
- Find/develop/justify prior information about the parameters in your model. They must be informative.
- Propose two alternative models based on adding/subtracting predictors or including interaction or higher-order terms, etc. These models do not need to have informative priors, although they can if you would like.
- Also fit a model with your original predictors but use the default, weakly informative priors in rstanarm.
- Evaluate and compare the four models and choose the “best” model.
- Summarize your chosen model. Include interpretations of posterior credible intervals for the beta coefficients in the model.
- For a set of covariates of interest, conduct a posterior predictive analysis, including an interval.

Assessment

- A project proposal with the data set, identifying the response variable and the predictors to be used in the initial model
- A paper summarizing your work
- An R script with your code
- A “poster” presentation for the final exam slot: one slide containing the important information from your project